

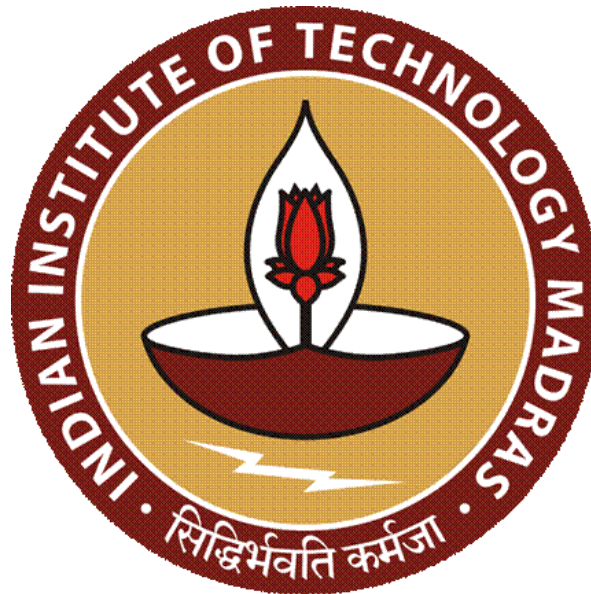
Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sales

A Mid Term report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sales”. I extend my appreciation to Navkar Enterprises, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate

Name: Vinshi Jain

Date: 1 February 2025

1. EXECUTIVE SUMMARY AND TITLE

The project, titled "**Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sales**" commenced with the acceptance of the proposal to streamline inventory management processes for NAVKAR ENTERPRISES. The goal is to enhance operational efficiency and reduce costs by employing data-driven strategies.

Descriptive Analysis highlighted critical insights such as the significant sales contribution of products, while also identifying wastage concerns for items like Adhesive, silicone. The enterprise's purchasing and sales records were thoroughly examined to understand current inventory levels, usage, and turnover rates, using tools like Excel and Python.

To focus efforts on the most impactful products, **Pareto Analysis** was used to identify the top 20% of product categories that generate around 80% of the revenue. This helped prioritize production and inventory management around the most profitable sticker types.

Additionally, **What-If Analysis** in Excel was applied to simulate various scenarios for sales and demand forecasts. This helped predict how changes in production or material usage might impact costs and inventory levels, enabling smarter decision-making.

Trend Analysis revealed steady sales patterns for customised stickers and logo stickers, with products like cartoon stickers showing more variability. This predictive insight has already helped inform purchasing strategies, ensuring that stockouts are minimized while overstocking perishable goods is avoided. Through these analyses, early findings suggest areas where better control can reduce wastage and improve cost management, with the potential to boost profitability and operational sustainability. The report will further expand on the methodology and insights, supporting the strategic approach being developed for NAVKAR ENTERPRISES.

2. PROOF OF ORIGINALITY

1. **Letter of Authentication from the Organization** : The letter can be accessed through the link: [NAVKAR ENTERPRISES](#)
2. **Images of the factory and godown** : [Images and videos](#)
3. **Gmeet with owner**:[google meet](#)

3. METADATA AND DESCRIPTIVE STATISTICS

I collected data from 1st October 2024 to 31st December 2024, tracking the daily inflow of product material, including wastage. The data was initially recorded manually in a traditional notebook, requiring transcription into Excel for more detailed analysis. As the shop uses outdated methods for data entry and management, some figures might be approximate or slightly inaccurate due to manual calculation errors.

3.1 METADATA:

1. Inventory Data :

- Date: Date of the inventory record (format: YYYY-MM-DD).
- Item: Name of the item in the inventory (for example, Ink, Silicone, Adhesive, Vinyl).
- Quantity Purchased: Number of units purchased .
- Cost per Unit: Price per unit of the item purchased.
- Total Cost: Total purchase cost, calculated as $\text{Quantity Purchased} \times \text{Cost per Unit}$
- Daily Usage: Average daily consumption of the item.
- Remaining Stock: Current stock of the item available in the inventory.

2. Sales Data:

- Date: Date of the sale.
- Product Type: Category of the product sold
- Item: Name of the sticker sold.
- Quantity Sold: Number of units sold (in pieces).
- Unit Price: Selling price per unit (₹ per piece).
- Cost Price: Cost price per unit (₹ per piece).
- Total Cost: Total cost price for the quantity sold.
- Total Revenue: Total sales revenue ($\text{Quantity Sold} \times \text{Unit Price}$, ₹).
- Profit: Profit from the sale ($\text{Total Revenue} - \text{Total Cost}$, ₹).

3. Waste Data:

- Month: Month for which the waste data is recorded (format: YYYY-MM).
- Item: Name of the item wasted (for example, Adhesive, Ink, Silicone, Vinyl).
- Total Quantity Purchased: Total quantity of the item purchased during the month (in kg).
- Quantity Wasted: Amount of the item wasted during the month (in kg).

4. Competitor Price Data:

- Competitor Name: Name of the competitor (for example, LabelX, FootMark Labels, InstaSticker).

- Product Type: Type of product for which the competitor price is recorded (such as Cartoon Stickers, Logo Stickers, Custom Design Stickers).
- Period: The month and year for which the competitor price is applicable.
- Region: The geographical region where the competitor's price is applicable (for example, South, West).
- Price: The price at which the competitor is offering the product in the specified period and region.

3.2 DESCRIPTIVE STATISTICS

Each of the data tables was explored using descriptive statistics. The result can be inferred from the following table.

Descriptive statistics of inventory data

	date	quantity_kg	unit_price	total_cost	daily_usage_kg	remaining_stock_kg
count	200	200.000000	200.000000	200.000000	200.000000	200.000000
mean	2024-11-13 12:07:12	56.865000	191.030000	11022.460000	14.275000	42.590000
min	2024-10-01 00:00:00	10.000000	102.000000	1221.000000	1.000000	5.000000
25%	2024-10-21 00:00:00	31.000000	143.750000	5732.250000	5.750000	22.000000
50%	2024-11-15 00:00:00	59.000000	188.000000	9945.500000	12.000000	40.000000
75%	2024-12-07 00:00:00	80.000000	233.500000	15348.000000	21.000000	60.250000
max	2024-12-31 00:00:00	100.000000	300.000000	29700.000000	48.000000	96.000000
std	NaN	27.601194	54.240569	6649.600239	10.490454	23.733273

This table summarizes 200 records, with an average quantity of 56.87 kg, unit price of 191.03, total cost of 11,022.46, daily usage of 14.28 kg, and remaining stock of 42.59 kg. The maximums are 100 kg quantity, 300 unit price, 29,700 total cost, 48 kg daily usage, and 96 kg remaining stock, while the minimums are 10 kg, 102, 1,221, 1 kg, and 5 kg respectively.

Descriptive statistics of sales data

	sale_date	units_sold	unit_price	cost_price	total_cost	total_revenue	profit	profit_margin_pct
count	600	600.000000	600.000000	600.000000	600.000000	600.000000	600.000000	0.0
mean	2024-11-15 13:04:48	1340.000000	11.388333	6.666667	9893.333333	16975.000000	7081.666667	36.236667
min	2024-10-01 00:00:00	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	NaN
25%	2024-10-25 00:00:00	500.000000	7.000000	4.000000	4000.000000	7500.000000	2000.000000	18.000000
50%	2024-11-15 00:00:00	1500.000000	12.000000	6.000000	7500.000000	14000.000000	4500.000000	36.000000
75%	2024-12-08 00:00:00	2000.000000	16.000000	9.000000	14000.000000	25000.000000	10000.000000	56.000000
max	2024-12-31 00:00:00	2500.000000	20.000000	19.000000	47500.000000	50000.000000	42500.000000	85.000000
std	NaN	804.972028	5.919961	4.442422	8598.799595	12729.045106	7439.516644	23.451167

This table summarizes 600 sales records, showing an average of 1,340 units sold, unit price of 11.39, total revenue of 16,975, and average profit of 7,082 with a mean profit margin of 36.24%¹. The maximum recorded values are 2,500 units sold, 20 unit price, 50,000 total revenue, and 42,500 profit, while the minimums are all zero.

Descriptive statistics of waste data

	total_quantity_purchased	total_waste_kg	waste_percentage	round off values
count	12.000000	12.000000	12.000000	12.000000
mean	947.750000	71.083333	7.200669	0.070000
std	293.627199	29.128113	2.021474	0.020889
min	411.000000	11.000000	2.676399	0.020000
25%	781.500000	61.250000	6.663060	0.067500
50%	995.000000	82.500000	8.032761	0.080000
75%	1079.750000	91.250000	8.266963	0.080000
max	1529.000000	105.000000	9.366637	0.090000

This table summarizes 12 records, showing an average total quantity purchased of 947.75 kg, total waste of 71.08 kg, and waste percentage of 7.20%. The maximum values are 1,529 kg purchased, 105 kg waste, and 9.37% waste percentage, while the minimums are 411 kg, 11 kg, and 2.68% respectively.

Descriptive statistics of competitor data

	price
count	27.000000
mean	12.370370
std	4.583819
min	4.000000
25%	9.000000
50%	13.000000
75%	16.000000
max	20.000000

This table summarizes 27 price records, with an average price of 12.37, a minimum of 4, a maximum of 20, and a median (50%) of 13. The prices have a standard deviation of 4.58, with 25% of values below 9 and 75% below 16.

Python file :

https://drive.google.com/file/d/1Hyrd0k18Z2ynoHJdlvJ1cUeOraLgMYLV/view?usp=drive_link

4.Detailed Explanation of Analysis Process/Method

Analysed Data: [SHEET](#)

The analysis aims to provide actionable insights to enhance operational efficiency, optimize inventory, and maximize sticker sales at NAVKAR ENTERPRISES. The data has multiple domains – inventory, sales, waste, and competitor sales data – across three months of 2024, forming a structured foundation for both descriptive and diagnostic analysis.

Analysis Process

- **Summary statistics** (mean, min, max, standard deviation) were calculated for all the sheets using Python.
- Monthly trends were visualized using **line charts** and **bar graphs** in Excel to reveal cyclicity in sales and demand.
- A **Pareto Analysis** (80/20 Rule) was applied using a cumulative percentage column to highlight the top 20% of products responsible for ~80% of revenue.
- A Pivot Table in Excel was used to summarize average usage and turnover per product per month.
- **Inventory Turnover Ratio** was calculated as: $\text{ITR} = \text{daily_usage_kg} / ((\text{quantity_kg} + \text{remaining_stock_kg}) / 2)$
- **Conditional formatting (color scale)** was applied in Excel to visually identify high-waste materials for cost control.
- A bar chart was used to compare price differences across months and competitors.

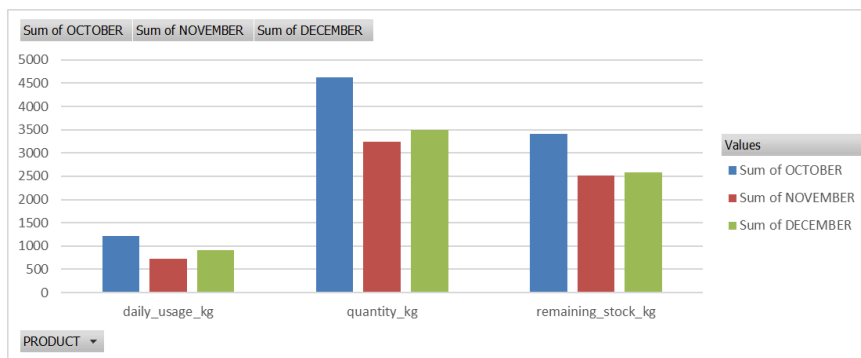
5. RESULTS AND FINDINGS

Metric	Value
 Highest Revenue	₹5,31,820
 Highest Profit %	95.02%
 Lowest Revenue	₹89,400
 Lowest Profit %	50.16%
 Average Revenue	₹2,94,170
 Avg. Profit %	74.52%

Findings based on business challenges :

1.STRUGGLES WITH INVENTORY MANAGEMENT

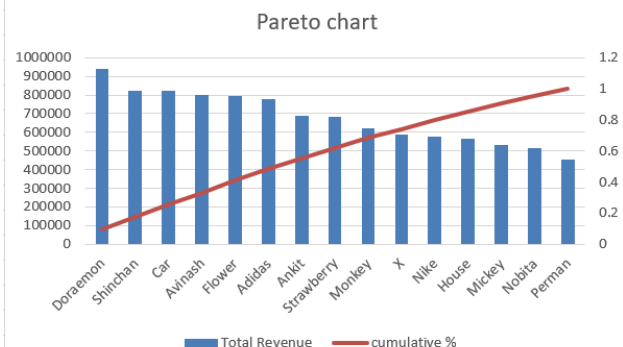
PRODUCT	OCTOBER	NOVEMBER	DECEMBER
quantity_kg	4631	3240	3502
AVG daily_usage_kg	1217	726	912
AVG remaining_stock_kg	3414	2514	2590

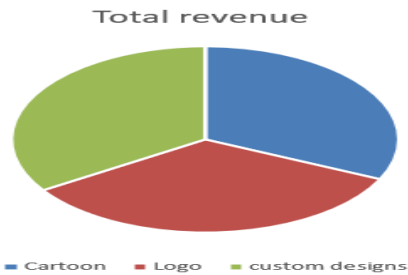


- October had the highest material purchase (around 4600 kg). There is a visible drop in November, and then a slight recovery in December. This could indicate bulk stocking in October, possibly anticipating higher demand or discounts from suppliers.
- Stock left over in October is also high (around 3400 kg), confirming that not all material was consumed. We should Track and forecast usage better to avoid overstocking like in October.

2.LACK OF CLARITY IN STOCK LEVELS AND USAGE

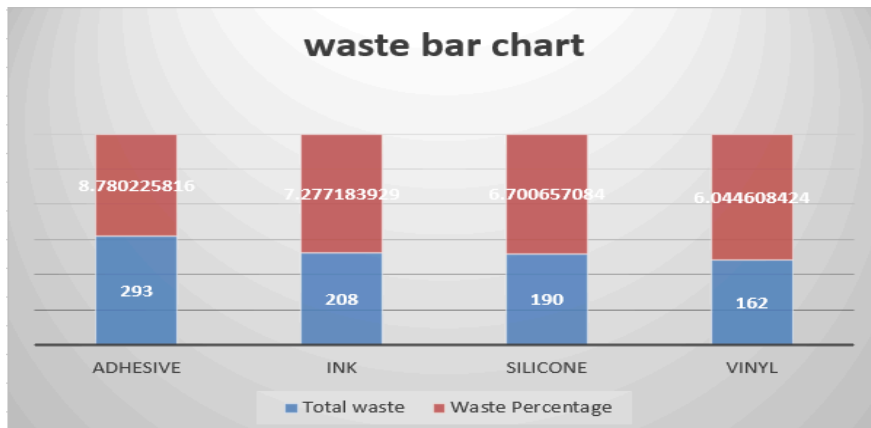
Sticker name	Total Revenue	cumulative %
Doraemon	938000	0.09209622
Shinchan	825500	0.173146784
Car	820500	0.253706431
Avinash	801000	0.332351497
Flower	795500	0.410456554
Adidas	778500	0.486892489
Ankit	690500	0.554688267
Strawberry	682000	0.621649485
Monkey	622000	0.682719686
X	588000	0.740451645
Nike	579000	0.797299951
House	564000	0.852675503
Mickey	533000	0.905007364
Nobita	516000	0.955670103
Perman	451500	1





- The stickers on the left (Doraemon, Shinchan, Car...) contribute the most to total revenue. The revenue gradually decreases as we move right (towards Mickey, Nobita, Perman).
The red line shows the proportion of total revenue covered cumulatively as we go from top-selling to lowest-selling stickers.
- The top 11 stickers (up to "House") account for 85% of the revenue, confirming the Pareto Principle, which says:
Roughly 80% of effects come from 20% of causes.
In this case a small group of stickers generates most of the revenue.

Material type	Total waste	Waste Percentage
Adhesive	293	8.780225816
Ink	208	7.277183929
silicone	190	6.700657084
Vinyl	162	6.044608424



Total Waste (Blue Bars)

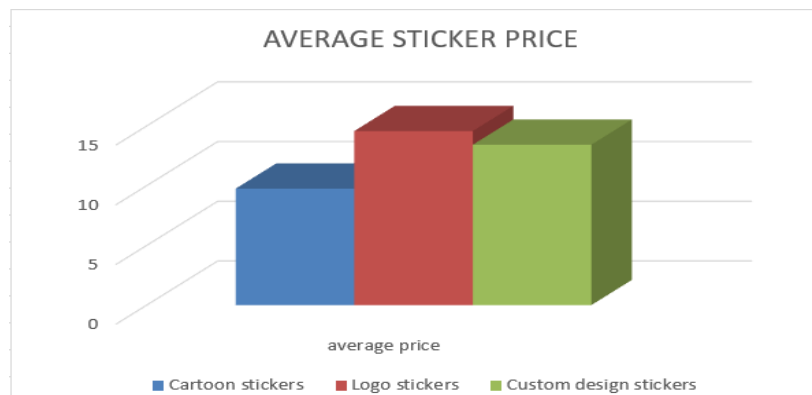
- Adhesive contributes the most to waste (293 kg).
- Vinyl has the least waste (162 kg).

Waste Percentage (Red Segment on Top)

- The waste percentage is also highest for Adhesive (8.78%) and lowest for Vinyl (6.04%).

RISING PERCENTAGE AND PRICE WARS

product type	average price
Cartoon stickers	9.75
Logo stickers	14.57142857
Custom design stickers	13.42857143



Logo Stickers have the highest average price:

- ₹14.57, suggesting they might involve branding, premium quality, or small-batch production.

Custom Design Stickers are next:

- ₹13.43 – These require personalization and effort, which explains their higher cost compared to cartoon stickers.
- ₹9.75 – Possibly mass-produced and sold in larger volumes or bundles.

Cartoon Stickers are the most affordable:

- ₹9.75 – Possibly mass-produced and sold in larger volumes or bundles.

